

Bharth K S

bharth.ks2003@gmail.com | +447304496854 | [LinkedIn](#) | Edinburgh, United Kingdom

Portfolio: [Click to view my Portfolio](#)

Summary

Applied AI Engineer with an MSc in Robotics, specialising in Generative AI and Agentic AI solutions. Expertise includes LLM integration, prompt engineering, adversarial safety testing, and RAG-based systems. Proficient in Python for building production-grade services and deploying solutions on cloud-native platforms. Seeks to apply strong software engineering skills to design, build, and productionise scalable AI/ML solutions.

KEY ACHIEVEMENTS

- Designed and deployed multi-LLM adversarial testing framework to evaluate AI robustness against prompt attacks.
- Fine-tuned transformer-based language models to improve safety, reliability, and response quality in AI agents.
- Developed embodied AI security solutions within AI2-THOR environments for responsible AI research initiatives.
- Integrated LLM-powered conversational capabilities into robotic systems for natural language interaction workflows.
- Built cloud-connected IoT architecture supporting secure real-time sensor ingestion, analytics, and visualization.
- Engineered autonomous navigation algorithms utilizing SLAM and Kalman filtering for intelligent robot movement.

CORE COMPETENCIES

- Generative AI & LLM Engineering
- AI Agent Development & Automation
- REST API Development & Integration
- Retrieval-Augmented Generation
- LLM Fine-Tuning & Model Optimization
- AI Safety & Responsible AI
- Adversarial AI Testing & Defense

CERTIFICATIONS & TRAININGS

- Workshop Implant / Industrial Training** – Mould Care Centre (Jan 2023).
- Practical C++: Learn C++ Basics Step by Step** – Udemy (Jun 2025).
- AI Tools Workshop** – Be10x (May 2025).

WORK EXPERIENCE

PRODUCTION ENGINEERING INTERN

June 2025 - July 2025

Hyundai Motors Pvt. Ltd.

Chennai, India

- Automated quality validation for robotic production systems using Python and CI/CD tools, maintaining consistent reliability across assembly line operations.
- Analysed manufacturing performance datasets to identify efficiency trends, preventing production line stoppages through continuous process optimisation.
- Developed automated testing suites for production software using version-controlled workflows and containerisation to improve code reliability and deployment security.

Startup Ecosystem Associate

Dec 2025 - Present

Venture Cafe

Edinburgh, United Kingdom

- Designed and deployed GenAI solutions using Python, implementing RAG architectures and prompt engineering to automate business workflows and improve decision-making processes.
- Implemented MLOps practices using AWS SageMaker and CI/CD pipelines to manage model versioning, automated testing, and deployment cycles for production-grade AI services.
- Built and monitored agentic AI systems, developing observability and logging frameworks to track system performance, latency, and reliability in cloud-native AWS environments.

EDUCATION

- MSc Robotics**, Heriot-Watt University | Edinburgh, United Kingdom (Present).
- Bachelor of Engineering (B.E.) in Mechatronics Engineering**, Kongu Engineering College | Erode, India (2025).

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, C, MATLAB.
- **Generative AI & LLMs:** OpenAI API, Anthropic Claude, Ollama, Hugging Face, Prompt Engineering, LLM Fine-Tuning, RAG.
- **Artificial Intelligence:** Machine Learning, AI Safety, Responsible AI, Adversarial Testing, Hallucination Mitigation.
- **Computer Vision:** OpenCV, Image Processing, Visual Perception Systems.
- **Cloud & IoT:** MQTT, Node-RED, ThingSpeak, ESP32, Cloud IoT Architectures.
- **Robotics:** ROS 1, ROS 2, SLAM, Sensor Fusion, Kalman Filtering, Autonomous Navigation.
- **Development Tools:** Git, Jupyter Notebook, Docker, VS Code, AI2-THOR.
- **Engineering Software:** SolidWorks, Autodesk CAD, Ansys, LabVIEW, RoboStudio.

VOLUNTEERING

- **Team Member**, Robotics Makers Society, Heriot-Watt University | Edinburgh, United Kingdom (2025 – 2026).
- **Secretary**, The Robotic Society, Kongu Engineering College | Erode, India (2024 – 2025).
- **Technical Member**, The Robotics Club, Kongu Engineering College | Erode, India (2023 – 2024).

KEY PROJECTS

ADVERSARIAL SAFETY IN EMBODIED AI – LLM ATTACK & DEFENSE SYSTEM

(2025 - Present)

Heriot-Watt University

Edinburgh, United Kingdom

- Architected adversarial attack and defense framework for LLM-powered agents operating within AI2-THOR ecosystems.
- Developed multi-LLM infrastructure integrating Anthropic Claude and Ollama models for robustness evaluation.
- Implemented prompt injection testing workflows to assess vulnerabilities across embodied AI environments.

GENAI INTEGRATION WITH ROBOTIC SYSTEMS

(2025 - Present)

Heriot-Watt University

Edinburgh, United Kingdom

- Integrated advanced language models with robotic platforms enabling natural language understanding capabilities.
- Developed context-aware interaction workflows supporting intelligent decision-making and robotic responses.
- Applied prompt engineering strategies to optimize behavioral control and task execution performance.

CLOUD-CONNECTED IOT DATA PIPELINE – ENVIRONMENTAL MONITORING SYSTEM

(2025 - Present)

Heriot-Watt University

Edinburgh, United Kingdom

- Built end-to-end IoT architecture supporting real-time environmental monitoring and cloud-based analytics.
- Integrated ESP32 devices with MQTT communication protocols for secure sensor data transmission.
- Developed Node-RED workflows enabling automated processing and orchestration of incoming data streams.

AUTONOMOUS NAVIGATION SYSTEM – ROS 2 ROBOTICS ALGORITHMS

(2025 - Present)

Heriot-Watt University

Edinburgh, United Kingdom

- Developed autonomous navigation framework utilizing SLAM algorithms for real-time mapping capabilities.
- Implemented Kalman Filter methodologies improving localization accuracy and sensor fusion performance.
- Engineered production-quality Python and C++ code following software engineering best practices.

Rescue Agent (Hackathon Project) Agents for Humans for food Distribution

(2025 - Present)

DevPost Hackathons

Edinburgh, United Kingdom

- Developed a web application for restaurant managers to notify drivers regarding leftover food redistribution using AWS Strand and Claude 3.5 Opus.
- Designed and executed an internal pilot evaluation using 3 driver datasets and 50 food-specification JSON datasets to ensure output accuracy.
- Built the user interface using Streamlit to facilitate seamless interaction between managers and delivery personnel.